# CS301 Database Management Systems

## Assignment 3: Transactions and Concurrency Control

Northfield Institute of Technology - 2026 Odd Semester - Dr. Kavita Mehta

**Submission deadline: on or before 13 November 2026, 11:59 PM**

Submit one PDF on CampusAI.

### Task

Q1. Explain the four ACID properties with one banking example each.

Q2. Given the schedule S: R1(A) W2(A) R2(B) W1(B), draw its precedence graph and say whether it is conflict-s

Q3. Show how two-phase locking (2PL) prevents the lost update problem for two transfers on the same account.

Q4. In 150-200 words, compare strict 2PL with timestamp ordering.

### Marking scheme (20 marks)

- ACID properties with correct examples (5 marks)
- Precedence graph and serializability verdict (5 marks)
- 2PL lost-update walkthrough (6 marks)
- Comparison of strict 2PL and timestamp ordering (4 marks)